FEDERAL SPECIFICATION

CHAIN ASSEMBLIES, TIRE: FOR

OFF-THE-ROAD VEHICLES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers various commercial tire chains for large, off-the-road tires. The tire chains are for use in applications requiring tire protection and increased traction in severe usage. The tire chains are intended for use on rubber tire earthmover, construction, materials handling, and special use vehicles.

1.2 Classification. The tire chains shall be of the following classifications for the intended off-the-road (OTR) tires.

1.2.1 Type and class. The tire chains shall be of the following types and classes, as specified (see 6.2):

Type PE	- Protection, link type tire chains:
Class 1C	- 2.2 inches (56mm) or less opening between chain links, chain mesh of diamond, circular, or rhombus patterns.
Type PT	- Protection, track type tire chains:
Class 1T	- Closed, track chain around outer circumference of tires; transverse, single-bar, grouser-shoe type.
Class 2T	- Closed, track chain around outer circumference of tire; transverse double-bar, grouser-shoe type.
Type T	- Traction, link type tire chains:
Class 4C	- 4.0 inches (102 mm) or less opening between chain links; tread-chain mesh of rhombus pattern.
Class 8C	- 8.0 inches (203mm) or less opening between chain links; tread-chain mesh of rhombus pattern.
Class 14C	- 14.0 inches (356mm) or less opening between chain links or cross-chains; tread chain of mesh, bias, or transverse cross-chain pattern, including tie links.
Class 100C	- 12.0 inches (305mm) or less spacing between cross-chains; tread chains of transverse cross-chain pattern.

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1.2.2 Service. Tire chains shall be of the following service application criteria, including rated tire chain tangential pull capacity on tire, as specified (see 6.2):

Service LD	- Light duty on light equipment; 5000 pounds minimum tangential pull.
Service MD	- Medium duty on light equipment; Light duty on medium equipment; 6000 pounds minimum tangential pull.
Service SD	- Severe duty on light equipment; Medium duty on heavy equipment; 9700 pounds minimum tangential pull.
Service HD	- Heavy duty on medium to heavy equipment; Light to medium duty on extra heavy equipment; 14,000 pounds minimum tangential pull.
Service ED	- Extreme duty on heavy equipment; Heavy duty on extra heavy equipment; 18,700 pounds minimum tangential pull.

Note. For type P chains, see table I for standard and optional service selection.

1.2.3 Tire chain size. The size of the tire chain shall be in accordance with its matching, intended tire size and as specified (see 6.2).

Intended tires shall be those listed in the Yearbook and Supplementary Tire Data as published by the Tire & Rim Association, Incorporated (T&RA), and special driver tires as recommended by the original equipment manufacturer.

1.2.4 Chain components ordered for replacement. Chain components ordered for replacement shall be of the types, classes, service, and sizes specified herein, and the quantities specified (see 1.2.1 through 1.2.3, 3.7, and 3.10 through 3.10.3.2) (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Federal Standards

FED. STD. NO. 123	- Marking for Domestic Shipment (Civilian Agencies).
FED. TEST METHOD STD. 151.	- Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Settle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications

MIL-T-704	- Treatment and Painting of Materiel.

Military Standards

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW Washington, DC 20036.)

American Welding Society (AWS):

Welding Handbook

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

Industrial Fasteners Institute (IFI):

IFI 100	- Prevailing-Torque Type Steel Hexagon Locknuts.
IFI 101	- Torque-Tension Requirements for Prevailing-Torque Type Steel Hexagon Locknuts.
IFI 103	- Mechanical Requirements for Steel Machine Screws.
IFI 107	- Prevailing-Torque Type Steel Hex Flange Locknuts.
IFI 108	- Grade and Source Identification Marking of Grades B and C Prevailing-Torque Type Hex Locknuts.
IFI 111	- Hex Flange Screws.

(Application for copies should be addressed to Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

Forging Industry Association:

Forging Industry Handbook

(Application for copies should be addressed to the Forging Industry Association, Illuminating Building, Cleveland, OH 44101.)

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Society of Automotive Engineers (SAE):

SAE Handbook

HS 39	- Construction and Industrial Machinery Handbook Supplement
J429	- Mechanical and Quality Requirements for externally Threaded Steel Fasteners
J683	- Tire Chain Clearances
J727	- Nomenclature-Tractor (Track Type)
J833	- Male Physical Dimensions for Construction and Industrial Equipment Design
J925	- Minimum Access Dimension for Construction and Industrial Machinery

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, NY 10001.)

Tire and Rim Association, Inc. (T&RA):

Yearbook
Supplementary Service Data

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

3. REQUIREMENTS

3.1 Description. The tire chains shall be a special purpose vehicle accessory, providing protection and traction when wrapped around large off-the-road (OTR) pneumatic tires on earthmover, construction, materials handling, and special use vehicles. The tire chains shall be commercial products, intended for severe operation. The tire chains shall be of the same design, construction, materials, and quality as the tire chains recommended as optional accessory equipment by the original equipment manufacturers of earthmover, construction and materials handling equipment, in their publications. Flexible, type PE, link chains shall be of the small opening mesh type, restricting penetration of tire damaging object. Rigid, type PT, track chains shall be of the hinged track type installed around a special, pneumatic driver tire. The track, including grouser shoes, links, and pins and bushings, shall be as defined in SAE J727. The track chain shall also prevent penetration of tire damaging objects, while also providing increased flotation, traction, and vehicle stability. Flexible, type T, link chains shall be of the larger opening mesh or of the X-bias or transverse (radial) cross chain types, effectively increasing traction of the OTR lug tires in off-road terrain. Each tire chain shall be furnished with all components to meet the performance and requirements specified herein. Each tire chain shall function as intended on both free rolling and driving wheels of the vehicles.

3.2 First article (preproduction model). The supplier shall furnish one or more tire chains as specified (see 6.2), for examination and test within the time frame specified (see 6.2), to prove, prior to starting production, that his production methods and choice of design detail will produce chains that comply with the requirements of this specification. Examination and tests shall be as specified in section 4, and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Material. Material shall be furnished as specified herein and shall be new and unused. Material not specified shall be selected by the supplier as being compatible with the intended use in accordance with the engineering and production standards of the heavy-duty, earthmoving construction, and materials handling equipment tire chain industry and shall be as set forth in industry publications.

3.4 Stress calculation. The free body stress of any loaded component of the tire chain shall not exceed 100 percent of the guaranteed ultimate strength of the material under the following conditions:

(a) Maximum rated tire loading per T&RA Yearbook or tire manufacturer's published rating.
(b) Maximum available tire ply rating or load range per T&RA Yearbook or the tire manufacturer's published rating, whichever has the higher value.
(c) Maximum speed not more than 10 mph.
(d) Maximum coefficient of friction between the tire chains and the ground surface assumed to be 1.0.
(e) Driving axle having zero percent slippage.
(f) Penetration of chained tire and wheel assembly at 20.0 percent of the loaded rolling radius of the tire.

Stress calculations based on the above interrelated conditions may be furnished for evaluation, where correlating test results are not available.

3.5 Safety. The safety characteristics of the chains shall meet the following minimum requirements:

(a) Eliminate concealed danger created by the supplier's design.
(b) Eliminate hazards to personnel and to property.
(c) Result in furnishing material having a correct safety factor.
(d) Human factors engineering characteristics shall be incorporated into the chains. The operating characteristics of the chains shall provide for installation of the chains by personnel ranging from the small man, clothed, through the large man, arctic clothed, in accordance with SAE J833 and SAE J925.

3.6 Performance. The tire chains shall perform as specified herein when installed on the intended equipment specified (see 6.2). When the intended equipment is unknown in detail, each tire chain shall perform when installed on its matched tire and wheel assembly with the T&RA or tire manufacturer's published maximum rated tire loading for the maximum available ply rating (or load range) at various working speeds up to at least 10 mph. Performance criteria excludes chain failures attributable to sudden death causes. The minimum tire chain performance shall be as follows:

(a) For type PE protective tire chains, the protective tire chain life in abrasive material (see 6.7) shall be not less than the following:

(1) Extremely abrasive -1200 hours, minimum.
(2) Medium abrasive -2700 hours, minimum.
(3) Low abrasive -5000 hours, minimum.

(b) For type PT protective track tire chains, the protective tire chain life in abrasive material (see 6.7) shall be not less than the following:

(1) Extremely abrasive - Sand slurry type, 500 hours, minimum; unsaturated materials, 1000 hours, minimum.
(2) Medium abrasive - 2700 hours to pin turn, minimum.
(3) Low abrasive - 4000 hours to pin turn, minimum.

(c) For type T tractive chains, the tractive chain life in mud and soft ground shall be not less than 3500 hours of operation.

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3.7 Design and construction. Each tire chain shall be as represented in the chain manufacturer's publications and shall be comparable to the chain manufacturer's:

(a) Engineering data publications or chain design drawings (whichever has the later date)
(b) Manufacturing specifications.
(c) Quality assurance (quality control) specifications.
(d) Testing procedures.

The manufacturer's publications shall be identical to those furnished by the chain manufacturer for use by the commercial equipment manufacturer for the selection and sale of tire chain options. Other proprietary specifications, tolerances, and test procedures shall be those exhibited by the chain manufacturer, in confidence, to authorized Government personnel in accordance with the provisions specified in the contract or purchase order (see 6.8 and 6.8.1). The maximum variations of tire chain components between identical chains shall not exceed plus 15 percent or minus 2 percent of the values in the manufacturer's specifications and drawings. The maximum variations of chain weight shall not exceed minus 2 percent of the manufacturer's specified nominal chain weight.

3.8 Rockwell hardness. The Rockwell hardness of all tire chain components shall be as follows, unless otherwise approved by the contracting officer:

(a)	Type PE	Hardness Number C-60, minimum.
(b)	Type PT	Hardness Number C-45, minimum; except that industry practice shall apply to grousers, links and track pins.
(c)	Type T	Hardness Number C-50, minimum.

3.9 Case depth. All tire chain components shall have a minimum case hardness depth as follows, unless otherwise approved by the contracting officer:

(a)	Type PE	Case depth 0.027 inch (0.7mm) minimum; except forged links case depth 0.039 inch 1.0mm.
(b)	Type PT	None required on grousers and links; industry practice on track pins.
(c)	Type T	Case depth 0.15 inch (3.8mm), min. for forged and stamped links and between 8 to 15 percent of chain link wire and rod diameters.

3.10 Tire chain, type, class, and service. The type, class, and service of the tire chains shall be as specified (see 1.2.1 and 1.2.2). The tire chains shall be suitable for their intended use on the applicable equipment. In accordance with the service application of the tire chains, the chains shall withstand the following tangential pulls, imposed in one area of the chain:

(a)	Service LD	5,000 pounds, minimum.
(b)	Service MD	6,000 pounds, minimum.
(c)	Service SD	9,800 pounds, minimum.
(d)	Service HD	15,000 pounds, minimum.
(e)	Service ED	20,000 pounds, minimum.

The above tangential pull values shall be representative of those values encountered, when the tire chain is installed on the vehicle and it snags an immovable obstruction in path of travel and the vehicle tractive effort capability is gradually brought up to the value specified. Tire chains intended for tire protection applications, shall provide a metal interface between the tire and the traveling surface, effectively preventing sharp and abrasive objects, such as rocks, stumps, and refuse, from penetrating the tire chain barrier and causing tire punctures, damage, or abnormal wear. Tire chains, intended for tire traction applications, shall provide increased drawbar pull up to the capability of the equipment, dependent on the soil conditions. Tire chain replacement components, specified herein, shall be identical and interchangeable with like components of the tire chain. All connecting hardware and instructions for installation by user personnel shall be included.

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3.10.1 Type PE, protection link chains. The type PE protection link chains shall be constructed of one outer straight link side tension chain, one inner and one outer side chain, and a center section chain mesh. The strength of the side tension chain shall be equal to or greater than industrial high-test, proof-coil chain of the same size. The side tension chains shall have the chain mesh radially secured at each sixth side chain link, as a minimum, around the tire chain circumference. The side tension chains shall slip through the chain mesh radial rings to facilitate equal tightening tension of the chains. The chain mesh shall be constructed of a network of not less than four, center strut links fastened to one ring, providing a circular or a rhombus pattern. In the tire tread area the chain mesh shall have a maximum expanded metal-to-metal opening not exceeding 2.2 inches (56 mm). In the separation area, the chain mesh sections shall be joined by rows of single-hook connector links across the tread and inner and outer side chains. A toggle bar with U-bolt toggle bar clamp shall be furnished on the outer, side tension chain to secure the other end of the tension chain and to provide chain adjustment. When specified (see 6.2), four heavy duty cable clamps shall be furnished in lieu of the toggle bar, where side obstructions could damage or loosen the toggle bar. The limits of the chain mesh construction shall be in accordance with table I. Unless otherwise specified herein, the type PE protection link chains shall be of HD1 style. When specified (see 6.2), the protection link chains shall be of an optional style selected from table I.

Table I. Type PE Protection Link Chains

Standard or Optional	Opt.	Opt.	Opt.	Std.	Opt.
Service (See 1.2.2) Application	MD Rough Service	SD Rough Service	HD2 Rough to Heavy Service	HD1 Heavy to Extreme Service	ED Heavy to Extreme Service
Equipment Size	Light	Light	Light to Medium	Medium to Heavy	Heavy to Medium
Tractive Effort Capacity Min.	6,000 lb. (2700 kg)	9,700 lb. (4400 kg)	14,000 lb. (6400 kg)	14,000 lb. (6400 kg)	18,700 lb. (8500 kg)
Type of Link	Stamped or Formed & Welded	Forged, Stamped or Formed & Welded	Forged or Formed & Welded	Forged or Formed & Welded	Forged or Formed & Welded
Link Ground Contact Area Min.	0.60 sq. in. (390 sq. mm)	0.60 sq. in. (390 sq. mm)	0.92 sq. in. (590 sq. mm)	1.7 sq. in. (1100 sq. mm)	3.3 sq. in. (2130 sq. mm)
Link Dimensions:					
-Length, Min.	2.18 in. (55 mm)	2.18 in. (55 mm)	2.37 in. (60 mm)	2.37 in. (60 mm)	3.34 in. (85 mm)
-Height, Min.	1.00 in. (25 mm)	1.00 in. (25 mm)	1.56 in. (40 mm)	1.34 in. (34 mm)	1.87 in. (47 mm)
-Thickness Min.	0.28 in. (7 mm)	0.28 in. (7 mm)	0.39 in. (10 mm)	0.75 in. (19 mm)	1.00 in. (25 mm)

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Table I. Type PE Protection Link Chains (cont'd)

Standard or Optional	Opt.	Opt.	Opt.	Std.	Opt.
Ring thickness, Min.	0.26 in. (6.5 mm)	0.31 in. (8 mm)	0.43 in. (11 mm)	0.43 in. (11 mm)	0.54 in. (14 mm)
Chain-to-Vehicle Clearance Required. Max. (SAE J683)	1.5 in. (38 mm)	1.5 in. (38 mm)	2.2 in. (56 mm)	2.0 in. (51 mm)	2.7 in. (69 mm)

The type PE tire protection link chains shall function to close up the open pattern through flexing and gathering at the tire's foot print contact area in the direction of travel, thus presenting a metal surface, with a minimum of voids between the tire and the ground surface. With the protection link chain in correct adjustment, the flexing and gathering in front of the tire surface shall be not less than 1 inch (25 mm) nor more than 2.5 inches (64 mm). The drape of the chains down over the tire side walls shall be great enough to eliminate any need for side tensioning devices, other than the side tension chains around the perimeter of the chain assembly.

3.10.1.1 Chain rupture replacement mesh. When specified (see 1.2.4), chain rupture replacement mesh sections, in the quantity specified, shall be furnished with each tire chain. Each section shall include a full complement of connectors, one set on each end of the replacement mesh section. The length of each replacement mesh section shall be not less than the overall circumferential cord length of a 90-degree-central angel of the intended size of tire. The mesh pattern shall follow the basic pattern of the tire chain.

3.10.1.2 Ring replacement assembly. When specified (see 1.2.4), ring replacement assemblies, in the quantity specified, shall be furnished with each tire chain. The ring replacement assembly shall consist of an open ring and hold plate.

3.10.1.3 Link replacement assembly. When specified, (see 1.2.4), link replacement assemblies in the quantity specified, shall be furnished with each tire chain. The link replacement assembly shall consist of open links, roll pin sleeves and roll pins.

3.10.2 Type PT, protection track chain. Type PT protection track chains shall be constructed of heat treated grouser shoes continuously installed around the tread circumference of the matching driver tire. Track shoes shall be individually replaceable and the tire chain shall be removable as a chain assembly. The minimum necessary clearance shall be provided at hinge points. The grousers shall extend beyond the tire side walls on both sides. Unless otherwise specified (see 6.2), one special pneumatic driver tire shall be furnished by the track chain supplier with each track chain. The driver tire shall be compatible with and matched to the intended equipment and shall be in accordance with the recommendations of both the track chain manufacturer and the manufacturer of the intended equipment. Class 1T track chains shall be equipped with single bar grouser shoes, providing increased traction. In addition, Class 1T track chains shall include provisions to restrict lateral slippage of the equipment. Class 2T track chains shall be equipped with double-bar grouser shoes, presenting the greatest abrasive resistant bar area to abrasive rock.

3.10.2.1 Track replacement section. When specified (see 1.2.4), track chain replacement sections, in the quantity specified, shall be furnished with each tire track chain. The length of each track replacement section shall be not less than the overall circumferential cord chain length of a 90-degree central angle of the intended size of tire. The track pins in both ends of each section shall be furnished and shall be of the replaceable master pin type.

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3.10.2.2 Grouser replacement shoes. When specified (see 1.2.4), grouser replacement shoes, in the quantity specified, shall be furnished with each tire chain. The replacement shoes shall include the two, end-side-bars with each single-bar grouser when class 1T track chains are furnished.

3.10.2.3 Link, bushing, and pin replacement assemblies. When specified (1.2.4), assembly sets of two side links with two bushings and two, loose, track pins, compatible for one grouser shoe, shall be furnished in the quantity of sets specified with each tire chain.

3.10.2.4 Replacement driver tire. When specified (see 1.2.3), replacement driver tires, in the quantity specified, shall be furnished with each tire chain. The driver tire characteristics as defined by the tire manufacturer's engineering data publication or tire contour drawing, tire load-deflection chart data, manufacturing specifications, quality control specification, and tolerances shall be as recommended by the track chain manufacturer for use with his track chain.

3.10.3 Type T, traction link chain. Type T traction link chains shall be constructed of straight link side tension chains, one on each side, and multiple cross chains, either in an inter-connected pattern or radially side to side. The strength of the side tension chains shall be equal to or greater than industrial high-test proof coil chain of the same size. The side tension chains shall have the cross chains radially secured at each fourth side link, as a minimum, around the circumference. The cross chains shall be spaced evenly around the circumference of the tire. The interval between cross chains at the chain junction area shall not exceed 110 percent of the average spacing of the other cross chains of the tractors link chain. End hooks and lock fasteners shall be readily closed and opened manually; however, these fasteners shall be safe from accidental opening or detachment under severe earthmover service conditions. The traction link chains shall function in soft and slippery terrain to increase the drawbar pull capacity of the intended equipment beyond that obtained by the tire tread lug.

3.10.3.1 Side chain replacements. When specified (see 1.2.4), full length side chain replacements, in the quantity specified, shall be furnished with each tire chain.

3.10.3.2 Cross chain replacements. When specified (see 1.2.4), cross chain replacements, in the quantity specified, shall be furnished with each tire chain. Each cross chain replacement shall consist of all the components of a repetative pattern in the interconnected cross chain, or one strand of the radial side-to-side cross chain as applicable to the chain design.

3.11 Finish. All type PE and type T link chains shall be bright tumbled, shall be coated, or shall have a rust resistant finish to protect them in storage and in transit prior to use.

3.12 Treatment and painting. The portions of type PT track chains normally painted in commercial practice shall be cleaned, treated, and painted in accordance with MIL-T-704, type B.

3.13 Workmanship. All workmanship of each tire chain shall be in accordance with engineering, manufacturing and production standards of the earthmoving and construction tire chain industry. The chains shall be free from workmanship deficiencies that could impair the operation of serviceability of the chains. All parts, components and assemblies of the chains, including bushings, links, pins, rings, stampings, welded parts, castings, forgings, and machined work, shall be clean and free from sand, dirt, fins, pits, sprues, scales, flux, and other harmful, extraneous material. External surfaces shall be free of burrs and sharp edges and corners, except when sharp edges and corners are functional.

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3.13.1 Metal welding. All welding shall be in accordance with the American Welding Society's Welding Handbook and as specified herein. Welds shall provide the maximum notch toughness and fatigue strength with a minimum of residual stresses. All welds made on high-stress members shall be made from core or heavy-coated electrodes, gas-shielded, or submerged arc. Fillet welds shall be provided, when necessary, to reduce stress concentrations. The surfaces to be welded shall be free from scale, paint, grease, and other foreign matter. When multiple layers of weld metal are required, or each layer, or layer after the third pass with multipass welding process, shall be thoroughly cleaned before depositing another layer. All welded parts and assemblies shall be free from cracks and other imperfections that may reduce the effectiveness of the part or assembly. After welding, all weldments, shall be free from slag (except mill scale), flux, weld spatter, and other factors which affect the quality of the weld. Butt-type joints shall be alined within 20, 10, and 7.5 percent of the thickness of the equal or thinnest members involved for metal thickness of 1/32, 1/4, and 1/2 inch minimums, respectively. Where different metals are welded to each other, filler metal having strength of the lower strength metal shall be used. When the same metals, or metals having similar properties are used, the deposited filler metal shall develop the full design strength of the particular weld joint or weldment. Mechanical properties of all welds shall be such that welds will transmit stress without failure or permanent deformation when subjected to any test specified in this specification.

3.13.2 Bolted connections. Hardware shall be in accordance with SAE and IFI standards. All capscrews, bolts, and nuts shall be tight and correctly torqued. All junctures and joints shall be fastened with high strength bolts with washers and nuts, hex or 12-point type. Bolts shall be not less than SAE J429, grade B, minimum. Integral or separate washers shall be of the flat hardened steel type, sized to spread the clamping loads such that brinnelling or embedding in the component members shall be prevented. Mating nuts shall be of the prevailing torque type in accordance with IFI Standards, including IFI 100, 101, 103, 107, 108, and 111. Fasteners shall be lubricated prior to wrenching, and installation tools shall be proofed at set intervals to insure constant and correct torque performance.

3.13.3 Castings. Casting shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for its intended use.

3.13.4 Forgings. Forged components shall be in accordance with the Forging Industry Association Handbook.

3.13.5 Machine work. All machine parts shall be manufactured to gage through the use of correct jigs and fixtures and shall permit interchangeability of like parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and regulations.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Inspection of chain components (see 4.3).
(b) Preproduction inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection of preparation for delivery (see 4.7).

4.3 Inspection of chain components. All tire chain components shall be tested before being assembled into tire chains. Each lot of components shall be identified after testing to indicate whether it passed or failed inspection.

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4.3.1 Sampling. Sampling shall be in accordance with MIL-STD-105. A lot shall consist of like components.

4.3.2 Tests. Samples selected in accordance with 4.3.1 shall be tested in accordance with column 2 of table III. Any sample failing one or more tests shall be considered defective. AQL shall be 1.0 percent defective.

4.4 Preproduction inspection.

4.4.1 Examination. Each preproduction tire chain shall be examined for the defects listed in table II. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each preproduction tire chain shall be subjected to the tests marked "X" in column 1 of table III. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the defects listed in table II. Any sample with one or more defects shall be considered defective. AQL shall be 1.0 percent defective.

4.5.3 Tests. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in column 3 of table III. Any sample failing one or more tests shall be considered defective. AQL shall be 1.0 percent defective.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table II.

Table II. Classification of defects

Defect Number	Defects
101	Material not as specified (see 3.3).
102	Safety not as specified (see 3.5).
103	Design and construction not as specified or tire chains do not conform to manufacturer's specifications or drawings (see 3.7).
104	Tolerances exceed limits (see 3.7).
105	Tire chains do not conform to the requirements for type, class, and service (see 3.10 thru 3.10.3.2).
106	Component parts of tire chains not as specified (see 3.10 thru 3.10.3.2).
107	Finish, treatment, and painting not as specified (see 3.11 & 3.12).
108	Workmanship not as specified (3.13 thru 3.13.5).
109	Stress calculations, or test data in error incomplete, or not as specified (see 3.4 and 4.6.1.1).
110	One or more components of the chain not certified as being from an accepted lot of components (see 4.3).

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4.6.1.1 Stress calculations. Stress calculations test data for the mesh, track, or cross chain assemblies of the tire chain and as transferred to the two side chain or link-and-pin assemblies, shall be furnished. The theoretical load shall be as given in the T&RA Yearbook or the tire manufacturer's published rating for the maximum rated tire loading of the applicable tire size and maximum available ply rating (or load range) of the tire for a speed of not more than 10 mph. The tire-chain-to-ground coefficient of friction shall be assumed to be 1.0 on a driven axle with zero percent slippage. The number of chain components involved between the tire and the ground shall be calculated with the chained tire and wheel assembly penetration at 20.0 percent of the loaded rolling radius of the tires. Free body stress in any load-carrying component of the tire chain in excess of 100 percent of the guaranteed ultimate strength of the material or tire chains not as specified in 3.4 shall constitute failure of this examination.

4.6.2 Tests. Tests shall be in accordance with table III. An "X" in a column indicates the test that shall be conducted.

4.6.2.1 Test conditions. The tests in 4.6.2.2 and 4.6.2.4 shall be conducted at an ambient air temperature of 77° F. plus or minus 15° F. and all specimens shall be stabilized at that temperature before testing. The application test in 4.6.2.5 and 4.6.2.6 shall be performed without shelter and at the climatic conditions existing at the place of test. The method of chain tightening, duty cycling, and loading shall result in imposing stresses that are equivalent to those experienced during normal operation on earthmoving, quarry, construction, and materials handling job sites applicable to the chains.

Table III. Test Schedule

Pre-pro.	Compo-nents	Qual. Conf.	Test	Test Para.	Req. Para.
1	2	3	4	5	6
-	X	-	Hardness	4.6.2.2	3.8
-	X	-	Case Depth	4.6.2.3	3.9
X	-	X	Chain dimensions	4.6.2.4	3.7 & 3.10 thru 3.10.3.2
X (see 6.2)	-	-	Application	4.6.2.5	3.4, 3.5, 3.6, 3.7, 3.10 thru 3.10.3.2, 3.13 thru 3.13.5
X	-	-	Tangential pull	4.6.2.6	3.10
X	-	X	Installation	4.6.2.7	Section 3

4.6.2.2 Hardness. To determine conformance to 3.8 the tire chain components shall be subjected to the Rockwell hardness test as specified in method 243 of FED. TEST METHOD STD. No. 151. Nonconformance to the tire chain manufacturers specification, drawings, or quality assurance documents, or nonconformance to 3.8 shall constitute failure of this test.

4.6.2.3 Case depth. To determine conformance to 3.9 the tire chain components shall be prepared by grinding on a suitable abrasive wheel or sawing one cross section, perpendicular to the axis of chain component. The test specimen shall be etched in a solution of one part chemically pure nitric acid and one part water by volume (specific gravity 1.217). Specimens shall be etched for 6 seconds. Width of dark ring revealed shall be measured microscopically at 20 magnifications and recorded as depth of case. Nonconformance to the tire chain manufacturers specifications, drawings, or quality assurance documents or nonconformance to 3.9 shall constitute failure of this test.

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4.6.2.4 Chain dimensions. Measure and record the tire chain dimensions. The length of a chain shall be determined by hanging the chain by the closed chain fastener, vertically from a pin, the diameter of which shall be not greater than the diameter of the chain wire. The length of the chain shall be considered as the dimension from the top of the supporting pin to the bottom of a pin inserted in the bottom of the last link in the chain at right angles to the longitudinal axis of the chain and parallel to the supporting pin. The length of the chain shall be measured to the nearest 1/4 inch (6.4 mm). Nonconformance to the tire chain manufacturer's specifications, drawings, or quality assurance documents, or nonconformance to 3.10 through 3.10.3.2 shall constitute failure of this test.

4.6.2.5 Application. When specified (see 6.2), test the tire chain for at least 200 hours in the intended application at an operating site approved by the contracting officer. Obtain two identical vehicles, and equip one with the tire chains and operate the other without chains. To establish a wear pattern base; measure, mark, and record the tires and chains on the test vehicle and the tires only on the comparison vehicle. Accurately weigh and record the weight of each of the chains and each of the tires on the two vehicles. Work the vehicles side-by-side, in their intended application, doing an equal degree of payload movement and swapping drivers at least every shift. During the test, both scheduled and unscheduled maintenance is authorized; however, complete records of tire and chain data along with the data on each vehicle shall be maintained. Dependent upon conditions, sudden death failures are to be expected and analyzed. Failure of the chains to provide increased tire performance as specified in 3.6 or nonconformance to 3.4, 3.5, 3.7, 3.10 through 3.10.3.2 or 3.13 through 3.13.5 shall constitute failure of this test.

4.6.2.6 Tangential pull. Install and adjust a tire chain on the intended vehicle. Allow sufficient working of the chains to achieve stabilization of the chain tension on the tire. Attach a load imposing device to the tire chain, such as a band through and around the mesh type tire chains and the cross link tire chains; or a hook type plate under the grouser of track type tire chains. Rotate the tire until the load imposing device is directly under the axle. Load the wheel assembly up to the single tire maximum weight for 10-mph service as specified in the T&RA Yearbook or tire manufacturer's published rating for the size of tire at maximum commercially available load range or ply rating. Immobilize the vehicle, lock the tire-chained wheel, and slowly imposed a tangential pull on the tire chain up to the intended service application specified in 3.10. Hold the tangential pull for at least 1 minute. If breakage occurs, repeat the test for at least 10 other locations on the same tire chain. Record pull at breakage. Breakage at two or more locations on a tire chain, or nonconformance to 3.10 shall constitute failure of this test.

4.6.2.7 Installation. Wrap the tire chain around its applicable inflated tire or a rigid mock-up identical in size to the inflated tire. Simulate or, for questionable tire chains, actually connect the tire chains to normal mounted tension. Inability to simulate or actually install the tire chain or nonconformance to section 3 of this specification shall constitute failure of this test.

4.7 Inspection of preparation for delivery.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

111. Boxes not as specified for level A or B.
112. Box closure and strapping not as specified for level A or B.
113. Strapping not zinc coated for level A.
114. Gross weight exceeds the weight limitations for level B.
115. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each set of tire chains with the replacement components, when specified, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional. Box closure and strapping shall be in accordance with appendix to the box specification. Strapping shall be zinc coated.

5.1.2 Level B. Each set of tire chains with the replacement components, when specified, shall be packed in a close-fitting box conforming to PPP-B-601, domestic type, grade B, style optional. As an alternate a close-fitting box conforming to PPP-B-636, type SF or CF, class weather resistant; variety, grade, and style optional may be used. The gross weight shall not exceed the weight limitation of the box specification. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.1.3 Level C. Each set of tire chains with the replacement components, when specified, shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.2.1 or 5.2.2, as specified (see 6.2).

5.2.1 Civil agencies. All marking shall be in accordance with FED. STD. No. 123.

5.2.2 Military agencies. All marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tire chains are intended for use by the Federal Government to provide tire protection or vehicle traction or both for earthmoving, construction, materials handling, and special vehicle equipment on large off-the-road rubber tires. When equipment is operated in material which causes rapid tire wear (life of tire 500 hours or less), cutting or chunking of the tread, damage to the tire carcass, or frequent sudden death failures, the use of protective tire chains, types PE or PT, should be used. Where tire protection coupled with a mild increase in traction is needed, the type PE, protection chains should be ordered for the specific equipment. However, when both tire protection and increased traction and flotation are needed, the type PT, protection track chains should be ordered for the specific equipment. The type T, traction chains provide varying degrees of increased tractive effort in soft terrain, mud, snow and ice.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, service, and size of tire chains required (see 1.2.1 through 1.2.3). If the size of chain has not been determined, specify tire chains to be compatible with the intended tires as listed in the T&RA Yearbook of size, ply rating or load range, O-T-R tire type code number (such as E-2 or L-5), and type of tread design. In addition, if capable of being determined, the make-and-model and name plate data of the intended vehicle should be specified (see 3.6), to insure chain, tire, and vehicle compatibility (see 6.6).

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(c) Names and quantities of replacement chain components required concurrently with each chain or with each set of chains for the wheels of the intended vehicle (see 1.2.4, 3.7, and 3.10 through 3.10.3.2). (NOTE. The subsequent order for replacement chain components for in-use chains should be discouraged, unless assurance of interchangeability and chain manufacturer and vehicle manufacturer approval of the replacement chain components on their chains and their vehicle is obtained).
(d) The number of preproduction model chains required (see 3.2).
(e) Time frame required for submission of preproduction model (see 3.2).
(f) When cable clamps shall be furnished in lieu of the toggle bar on type PE chains (see 3.10.1).
(g) When type PE protection link chains of optional service MD, SD, HD2, or ED are required in lieu of the standard service HD1 (see 3.10.1).
(h) When special pneumatic driver tire for type PT protection track chains is not required with each chain (see 3.10.2).
(i) When application testing of the tire chains on a vehicle under similar usage conditions is required (see table III and 4.6.2.5).
(j) Level of packing required (see 5.1).
(k) Whether civil or military marking is required (see 5.2).

6.3 Preproduction model. Any changes or deviations of production chains from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish chains conforming to this specification.

6.4 Technical publications to accompany equipment. The contracting officer should furnish a list of technical publications as are required to be furnished.

6.5 Repair parts and maintenance tools. The contracting officer should furnish a list of repair parts and maintenance tools as are required to be furnished. No shipment of chains shall be made unless repair parts, maintenance tools, operating tools, and accessories are included with the shipment or unless approval for shipment of the chains without such items has been received from the contracting officer.

6.6 Tire chain options. Tire chain Government options, which may be exercised in 6.2 and are commercially available, are tabulated in table IV to facilitate applying the tire chains to the intended usage. An "X" in a column indicates that the type, class, and service is commercially available.

Table IV. Tire Chain Commercial Availability

Type and Class	Service ED	Service HD1	Service HD2	Service SD	Service MD	Service LD
Type PE (link)						
-Class 1C	X	X	X	X	X	-
Type PT (track)						
-Class 1T	X	-	-	-	-	-
-Class 2T	X	-	-	-	-	-
Type T (link)						
-Class 4C	X	X	X	X	X	-
-Class 8C	-	X	X	X	X	-
-Class 14C	-	X	X	X	X	-
-Class 100C	-	X	-	X	-	X

6.7 Definitions. The following are definitions of highly abrasive materials:

(a) Extremely abrasive. The following materials are examples of extremely abrasive materials: Granite having a very high quartz content; granit-based copper, zinc, and lead ores, taconite; molybdenum quartzite; and marbleized granite.

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(b) Medium abrasive. The following materials are examples of medium abrasive medium abrasive materials: Iron ore (not in taconite formation): slag; limestone or dolomite based copper, zinc, and lead ores, decomposed granite; limestone; and sandstones.

(c) Low abrasive. The following materials are examples of low abrasive materials; pure limestone and dolomite, shale and coal, gypsum, borax, and coral.

6.8 Proprietary information and data. The contracting officer should insure that the following clause is included in all invitations for bids or request for proposal, to insure that each chain manufacturer's proprietary information and technical data is safeguarded on a "For Official Use Only" basis:

"The proprietary manufacturing specification, quality assurance specifications, quality assurance tolerances, test prodedures, and any other proprietary information and technical data of the chain manufacturer, as required in accordance with 3.7, shall be exhibited by the chain manufacturer, in confidence, to authorized Government personnel on a classified material basis, solely for the purpose of engineering and quality assurance in connection with this contract or purchase order. This data shall be used within the chain manufacturer's establishment and shall not be divulged in any manner by authorized Government personnel, other than 'passed' or 'failed' criteria, without direct written permission of the chain manufacturer."

6.8.1 Subcontracts. The contracting officer should arrange for source inspection by the Government for equipment suppliers on whose items of equipment these chains and chain components are furnished (see 3.7).

MILITARY INTEREST:

Custodian:

Army - ME

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
USDA - AFS
TVA - TVA

PREPARING ACTIVITY: ARMY-ME

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to: General Services Administration, FSS,(FMA)Wash, DC 20406

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

RR-C-1832 CHAIN ASSEMBLIES, TIRE: FOR OFF-THE-ROAD VEHICLES

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA DC 69-9761

FORM GSA FED 69 2200

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